

MA 561: Fluidics and Microfluidics

Course Code: MA 561

Course Name: Fluidics and Microfluidics

L-T-P-C: 3-0-2-4

Prerequisites: MA 220 (PDE), MA 221 (Numerical Analysis) or equivalent

Intended for: BTech Maths and Computing 3rd year, 4th year, MSc, PhD, Elective for other disciplines.

Mutual Exclusion: ME 507

Elective or Core: Elective

Approval: 61st BoA

Preamble

The objective of this course is to provide students with a thorough understanding of fluid dynamics from fundamentals to microscale applications. Emphasis is placed on developing computational proficiency using COMSOL Multiphysics, a modern simulation platform widely used in research and industry, to model and analyze microfluidic, interfacial, and electrohydrodynamic flow phenomena.

Course Modules

- **Fundamentals of Fluid Dynamics:** Review of fluid properties: viscosity, density, surface tension, compressibility; Control volume and differential formulations; conservation of mass and momentum; Reynolds transport theorem; Navier–Stokes and Energy equations (steady and unsteady forms); Pressure-driven flows: exact solutions for Couette and Poiseuille flows; Stokes drag on a sphere, creeping flow, laminar flow approximations. (12 Hours)
Lab: Basics of Finite Difference; Numerical modelling of 1D pressure driven flows such as Couette flow, Poiseuille flow, Hagen- Poiseuille flow.
- **Introduction to Microfluidics and its applications:** Origin, definition, motivation, benefits, challenges, and applications; Physics of miniaturization, dimensional analysis, and scaling laws; Continuum hypothesis; Boundary conditions: slip theory, entrance effects, low Re regimes; Applications: Drug delivery and diagnostics; Biosensing and lab-on-a-chip systems. (10 Hours)
Lab: An overview of solving Navier-Stokes via any discrete numerical methods (understanding the sensitivity of pressure); Basics of COMSOL Multiphysics: model building, geometry, physics, mesh, study, post processing (result plotting 1D, 2D, and 3D). Fluid flow module- Flow past a sphere.
- **Capillary Flows:** Surface tension and interfacial energy; Young–Laplace equation, contact angle, capillary rise and capillary length; Interfacial boundary conditions; Marangoni effect; capillary flows. (10 Hours)
Lab: COMSOL: Laminar two-phase flow: (level set, phase field) Young–Laplace droplet on a flat substrate (2D axisymmetric); Capillary rise in a tube (2D axisymmetric); Marangoni-driven flow along an interface (thermocapillary and solutal marangoni).
- **Electrohydrodynamics & Microchannel Flows:** Fundamentals of electrohydrodynamics and species transport; Poisson–Boltzmann equation, Debye layer, thin EDL approximation; Electroosmosis: ideal Electroosmotic Flow (EOF), EOF with back pressure; EO devices- micropumps, micromixers, microvalve; EOF of power-law fluids. (10 Hours)
Lab: COMSOL: Lid driven cavity combined with mass transport and heat transport, Electric field in

a dielectric or leaky dielectric fluid, Electrostatic deformation of a droplet (leaky dielectric model), Electroosmotic flow in a microchannel, Electroosmotic flow in a microchannel connected reservoir, EOF of power-law fluids.

Laboratory/Practical/Tutorial Modules

Mentioned with the corresponding theory units.

Text Books

1. Bruus, Henrik. *Theoretical microfluidics*. Vol. 18. Oxford university press, 2007.
2. White, Frank M. *Fluid mechanics*. International Edition, McGraw-Hill (2015).

References

1. Kirby, B. J. – *Micro- and Nanoscale Fluid Mechanics: Transport in Microfluidic Devices*, Cambridge University Press.
2. Tabeling, P. – *Introduction to Microfluidics*, Oxford University Press.
3. Nguyen, N.-T. & Wereley, S. T. – *Fundamentals and Applications of Microfluidics*, Artech House.
4. Batchelor, George Keith. *An introduction to fluid dynamics*. Cambridge university press, 2000.
5. Chaurasia, Ashish S. *Computational fluid dynamics and comsol Multiphysics: A step-by-step approach for chemical engineers*. Apple Academic Press, 2021.
6. Multiphysics, COMSOL. “Introduction to comsol multiphysics®.” *COMSOL Multiphysics, Burlington, MA, accessed Feb 9.2018* (1998): 32.